CENTRAL DRY ZONE AGRICULTURAL SOIL LAB, TUMAKURU

SOIL HEALTH & FERTILITY TEST CERTIFICATE

SAMPLE DATA — FOR PROJECT DEMONSTRATION ONLY (FICTIONAL TEST DATA)

Soil Sample ID:	DEMO-SHC-KA06	**Report Date:**	20-Oct-2026
Location / District:	Tiptur Taluk, Tumakuru District, Karnataka	**Analytical Scheme:**	ICAR-NBSS&LUP; Standard Protocol
Soil Classification:	Coarse Red Sandy Loam	**Target Agro-Zone:**	Karnataka - Tumakuru Region

SOIL CHEMICAL AND PHYSICAL PARAMETERS

Parameter Name	Symbol	Observed Value	Unit	Reference Range	Rating / Interpretation
Available Nitrogen	**N**	**135.0**	kg/ha	280.0 - 560.0	Very Low (Severe Deficit)
Available Phosphorus	**P**	**21.0**	kg/ha	23.0 - 56.0	Low (Deficient)
Available Potassium	**K**	**155.0**	kg/ha	140.0 - 330.0	Low-Medium (Marginal)
Soil Reaction (pH)	**pH**	**7.10**	pH scale	6.5 - 7.5	Neutral (Optimal)
Electrical Conductivity	**EC**	**0.41**	dS/m	< 1.00	Normal (Non-Saline)
Organic Carbon	**OC**	**0.35**	%	0.50 - 0.75	Low (Poor Humus)

STANDARD ANALYTICAL READOUT (EXTRACTABLE KEY-VALUE STREAM):

Available Nitrogen (N): 135.0 kg/ha [Very Low (Severe Deficit)]

Available Phosphorus (P): 21.0 kg/ha [Low (Deficient)]

Available Potassium (K): 155.0 kg/ha [Low-Medium (Marginal)]

Soil Reaction (pH): 7.10 [Neutral (Optimal)]

Electrical Conductivity (EC): 0.41 dS/m [Normal (Non-Saline)]

Organic Carbon (OC): 0.35 % [Low (Poor Humus)]

SOIL HEALTH INTERPRETATION & NUTRIENT ADVISORY

Diagnostic Assessment: Coarse sandy soil prone to rapid moisture percolation and nutrient leaching. Low organic carbon reserve limits biological activity despite neutral pH reaction.

General Nutrient Management Guidance: Apply coir pith compost or tank silt @ 15 t/ha to elevate moisture holding capacity. Split nitrogen applications into 3-4 small fertigation doses using neem-coated urea. Well suited for coconut intercropping, ragi, and drought-tolerant pulses.

Soil Analyst: Demonstration Technical Officer

Quality Authorized by: Senior Agricultural Chemist

Status: Fictional Demonstration Data

ACADEMIC DEMONSTRATION DISCLAIMER: *This soil test report contains fictional demonstration data generated strictly for research, user interface validation, and OCR testing in the MannuMitra AI-Based Predictive Farming System. It is not an official laboratory certificate, does not represent physical soil sampling of any real farming parcel, and must not be used as official government documentation or for commercial agricultural transactions.*